

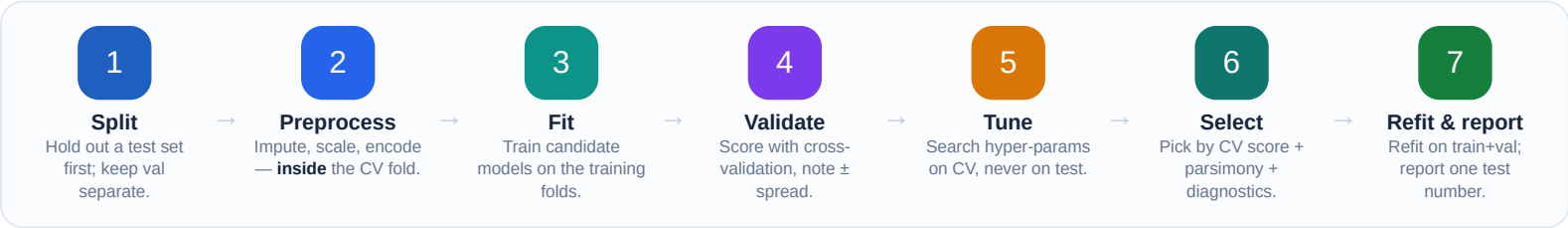
Build the Best Model: Workflow, Overfitting & Validation

The end-to-end path from raw data to a model you can trust — and the three numbers that tell you if it learned the signal or memorized the noise.

THE ONE RULE

The only score that estimates performance on brand-new data is cross-validated. Training and holdout are diagnostics — the CV mean ± spread is the truth.

THE END-TO-END MODELLING PIPELINE



🔒 Splitting & leakage guard

- ▶ **Train / Validation / Test** ≈ 60 / 20 / 20. Touch test **once**, at the very end.
- ▶ **Stratify** on the label for classification so each split keeps class balance.
- ▶ **GroupKFold** when rows share a source (same slab/patient/user) — split by the group.
- ▶ **TimeSeriesSplit** for temporal data — train on the past, test on the future only.
- ▶ Fit scalers, imputers, encoders, and feature selection **inside** each fold (use a Pipeline).

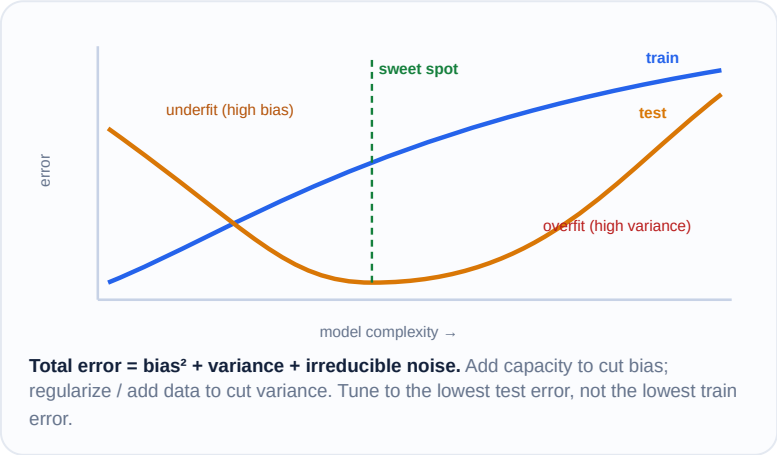
Leakage sources: target leakage (a feature encodes the answer), train → test contamination (preprocessing fit on all data), and temporal leakage (future info in past rows).

🔑 The three R² — read them together

SCORE	WHAT IT IS	TRUST
Training R ²	Fit on data the model has already seen.	Diagnostic only
Holdout R ²	One untouched 20% split.	Quick check
Cross-val R ²	Averaged over k held-out folds, with ± spread.	The headline

Overfit: high train, big drop to CV, wide ±. **Underfit:** low everywhere. **Healthy:** all close & high, tight ±.

BIAS-VARIANCE & THE COMPLEXITY SWEET SPOT



Total error = bias² + variance + irreducible noise. Add capacity to cut bias; regularize / add data to cut variance. Tune to the lowest test error, not the lowest train error.

📈 Learning & validation curves

- ▶ **Learning curve** (score vs training-set size): gap that stays wide → variance, get more data / regularize. Both curves low & converged → bias, add capacity / features.
- ▶ **Validation curve** (score vs one hyper-parameter): find where CV peaks before it turns down.
- ▶ **Early stopping:** halt boosting / neural-net training when the validation score stops improving.

CROSS-VALIDATION SCHEMES — PICK THE ONE THAT MATCHES YOUR DATA

SCHEME	HOW IT WORKS	USE WHEN
k-Fold typically k = 5 or 10	Rotate each fold as the test set once; average the k scores.	Default for i.i.d. tabular data.
Stratified k-Fold	k-Fold that preserves class proportions in every fold.	Classification, especially imbalanced.
GroupKFold	No group appears in both train and test.	Repeated measures / clustered rows.
TimeSeriesSplit	Expanding window; always predict forward in time.	Any temporal / forecasting problem.
Repeated k-Fold	Run k-Fold several times with new shuffles; average.	Small data — tighten the estimate.
Nested CV outer + inner loops	Inner loop tunes hyper-params; outer loop measures the tuned model.	Honest score and tuning at once.

HYPER-PARAMETER SEARCH & CAPACITY CONTROL

🔧 Tuning strategies

- ▶ **Grid search** — exhaustive over a small, sensible grid.
- ▶ **Random search** — sample the space; wins when few params matter.
- ▶ **Bayesian optimization** (Optuna / Hyperopt) — model the score surface, spend trials where they help.
- ▶ **Successive halving** — start many configs cheaply, keep promoting the best.

Always tune against the **CV score**; keep the test set sealed.

🔴 Regularization (shrink to fight variance)

Ridge (L2): $\min \sum (y - \hat{y})^2 + \lambda \sum \beta_i^2$ — shrinks coefficients smoothly.

Lasso (L1): $\min \sum (y - \hat{y})^2 + \lambda \sum |\beta_i|$ — drives some β to 0 (feature selection).

Elastic Net: mixes L1 + L2 ($\lambda[\alpha|\beta|_1 + (1-\alpha)|\beta|^2]$).

Trees / NN: limit `max_depth`, `min_samples_leaf`, learning rate, subsample, dropout, and use early stopping.



TAKEAWAY

Split honestly, validate with cross-validation, tune on the CV score, then **refit and report one sealed test number**. Fit and generalization are read together — never a single score.

Regression Metrics: Meaning, Formula & Acceptance Range

Every number the target is continuous — what it measures, its equation, and the range that counts as good. Match the metric to the cost of the error and the target's scale.

ERROR METRICS — SIZE OF THE MISS

Metric	Formula	Meaning & When to Use	Target
MAE Mean Absolute Error	$(1/n) \sum y_i - \hat{y}_i $	Average miss in the target's own units; robust to outliers. Report to non-specialists.	lower; 0 = perfect
MSE Mean Squared Error	$(1/n) \sum (y_i - \hat{y}_i)^2$	Punishes big misses quadratically; the quantity most models optimize.	lower
RMSE Root Mean Squared Error	$\sqrt{[(1/n) \sum (y_i - \hat{y}_i)^2]}$	MSE back in target units; sensitive to large errors. Compare vs σ of y .	lower; < $\sigma(y)$
RMSLE Root Mean Sq. Log Error	$\sqrt{[(1/n) \sum (\ln(1+\hat{y}) - \ln(1+y))^2]}$	Penalizes relative error; good for skewed / multiplicative targets, under-prediction.	lower
MAPE Mean Abs. % Error	$(100/n) \sum (y_i - \hat{y}_i)/y_i $	Scale-free % error; fails if $y \approx 0$. Common in forecasting.	<10% good
sMAPE symmetric MAPE	$(100/n) \sum 2 \hat{y}-y / (y + \hat{y})$	Bounded, symmetric version; handles small y better.	lower
MedAE Median Absolute Error	$\text{median}(y_i - \hat{y}_i)$	Fully outlier-robust typical miss.	lower
Huber Huber loss (δ)	$\frac{1}{2}a^2 \text{ if } a \leq \delta; \delta(a - \frac{1}{2}\delta) \text{ else}$	Quadratic near 0, linear in the tails — MSE accuracy with MAE robustness.	lower

GOODNESS-OF-FIT — SHARE OF VARIANCE EXPLAINED

Metric	Formula	Meaning	Range
R² Coefficient of Determination	$1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$	Fraction of variance explained. Rises with any added predictor — never judge alone.	$(-\infty, 1]$
Adjusted R²	$1 - \frac{(1 - R^2) \cdot (n - 1)}{(n - p - 1)}$	R ² penalized for predictor count; use to compare models of different size.	$\leq R^2$
Explained Var.	$1 - \frac{\text{Var}(y - \hat{y})}{\text{Var}(y)}$	Like R ² but ignores systematic bias in the residuals.	$(-\infty, 1]$
Acceptance is field-dependent: physical/engineering fits often expect R ² > 0.9; noisy human-behaviour or social data may be strong at R ² $\approx$ 0.3. Always compare against a baseline model.			

MODEL SELECTION & PARSIMONY — FIT VS COMPLEXITY

Criterion	Formula	Rule
AIC Akaike Information Criterion	$2k - 2 \ln(\hat{L})$	Lower is better. k = #parameters, $\hat{L}$ = max likelihood. $\Delta AIC > 10$ = decisive.
BIC Bayesian Information Criterion	$k \cdot \ln(n) - 2 \ln(\hat{L})$	Heavier complexity penalty than AIC → favours simpler models on big n .
Mallows' Cp	$RSS_p / \sigma^2 - n + 2p$	Good subset when $C_p \approx p$. RSS = residual sum of squares.
F-statistic	$(\text{explained}/p) / (\text{residual}/(n-p-1))$	Tests whether the model beats an intercept-only fit (overall significance).
Log-likelihood $\ln(\hat{L})$	$\sum \ln f(y_i \text{model})$	Higher = better fit; the raw input to AIC / BIC.

RESIDUAL DIAGNOSTICS — IS THE FIT EVEN TRUSTWORTHY?

Check	Formula / Test	Flag & Action
VIF Variance Inflation Factor	$VIF_j = 1 / (1 - R_j^2)$	> 5–10 → multicollinearity; drop/combine predictors or use ridge.
Condition number	$\sqrt{\lambda_{\max} / \lambda_{\min}}$	> 30 → unstable coefficients; center/scale, remove redundancy.
Durbin–Watson	$\frac{\sum (e_t - e_{t-1})^2}{\sum e_t^2}$	Range 0–4; ≈ 2 ideal. <1.5 or >2.5 → autocorrelation (order/robust SE).
Breusch–Pagan	χ^2 test on residual variance	$p < 0.05$ → heteroscedasticity; use robust SE or transform y .
Cook's distance	$D_i = (r_i^2/p) \cdot h_i / (1 - h_i)$	$D_i > 4/n$ → influential point; inspect, don't blindly delete.
Jarque–Bera normality of residuals	$(n/6)(S^2 + \frac{1}{4}(K-3)^2)$	Over-rejects on big n — judge by skew ≈ 0 , kurtosis ≈ 3 , and Q–Q plot.



TAKEAWAY

Report an **error metric in real units** (RMSE/MAE) with **R^2 / adjusted R^2** , choose between models on **AIC/BIC**, and confirm the residual diagnostics before you believe a single coefficient.